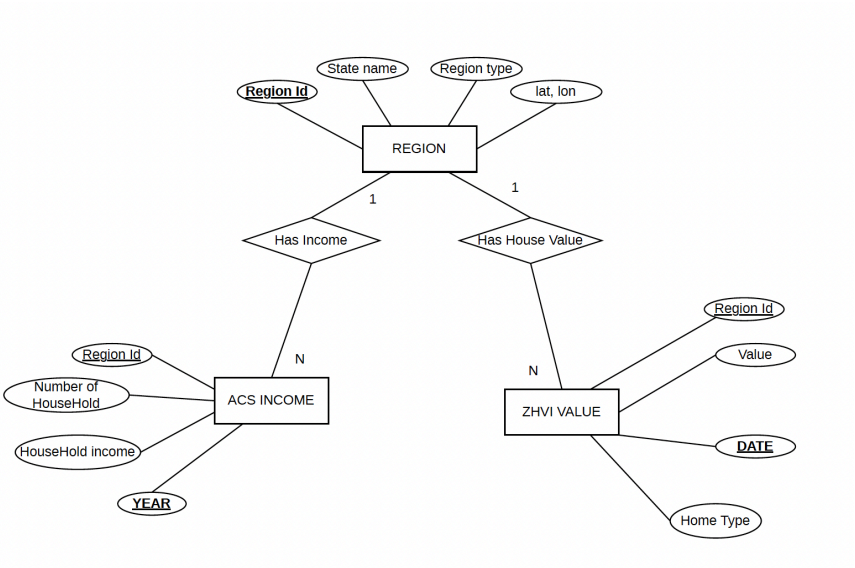
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Group 13
CIS 5500 Milestone 2
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1. This application aims to empower users with the knowledge to make well-informed decisions regarding the affordability of homes in areas. Someone who is considering moving, making a career change, or purchasing a home will benefit from data and insights on a particular region or city prior to making the life-altering decision to settle there.
2. List of planned features:
    - Search for a specific region or city to view trends in home value/income over time
    - Rankings for ZHVI
    - Rankings for ACS Census Data
    - Rankings for affordability ratio based on joined data
    - Income vs ZHVI interactive plot
3. List of features you might implement in the application, given enough time:
    - Log in page - save searches/create user profile to find regions that match user's income.
    - Region comparator - Compare two regions side-by-side
    - Region similarity - this will do clustering to find regions that are similar to a given region
    - Deployment to AWS ec2/rds
    - Integrate job posting dataset?
    - Live data from zillow?
4. List of pages:
    - Home Page/Search Page: This page will display a chart which plots the latest ZHVI and ACS income data in two axes to visualize where regions exist compared to each other. With this display, a user can click on or search for a specific data point to navigate to that region's page.
    - Region page: This page will display summary statistics/charts pertaining to the affordability ratio, ZHVI, and ACS income data of a specific region/city. It will highlight how these values have trended over time.
    - ZHVI Rankings: This page will show the top/bottom 25 ranked regions based on the ZHVI data. The user will be able filter this data to select regions of interest.
    - Income rankings: This page will show the top/bottom 25 ranked regions based on the ACS income data. The user will be able filter this data to select regions of interest.
    - Affordability ratio rankings: This page will show the top/bottom 25 ranked regions based on the joined ZHVI and ACS data. The user will be able filter this data to select regions of interest.
    - Log-in page?
5. Relational schema as a single ER diagram.



6. SQL DDL for creating the database.

```
CREATE TABLE region (
        region_ID  AS Primary KEY,
        state_name text,
        region_type text,
        lat double precision,
        lon double precision
);

CREATE TABLE acs_income (
        region_ID int,
        year int,
        household_income double precision,
        num_households int,

        PRIMARY KEY (region_ID, year),
        FOREIGN KEY (region_ID) REFERENCES region(region_ID)
);

CREATE TABLE zhvi_value (
        region_ID int,
        date DATE,
        value double precision,
        home_type text,

        PRIMARY KEY (region_ID, date),
        FOREIGN KEY (region_ID) REFERENCES region(region_ID)
);
```

7. Explanation of how you will clean and pre-process the data:
    - Standard cleaning (remove/replace nulls, drop unnecessary columns, normalize dates etc)
    - Pivot ZHVI dataset to use long shape for time-series data
    - Normalize geographic region names for joining
    - Inflation adjusted dollar amounts?

8. List of technologies you will use. You must use some kind of SQL database. We recommend using PostgreSQL or Oracle specifically because we will provide guidance for setting up a PostgreSQL database (example can be found in Homework 1 Assignment posted on Canvas):
    - Frontend: React
    - Backend/API: NodeJS or FastAPI
    - Database: Postgres

9. Description of what each group member will be responsible for:
    - Frontend:
        - Create each page/frontend skeleton - Quinton
        - Create charts/visuals - Quinton
        - Implement frontend logic - Quinton
    - Backend:
        - Create routing/api: Vaughan
        - Write initial queries - Aiden
        - Query Optimization - Aiden
        - Populate DB: Vaughan
    - Cleaning/pre-processing:
        - Create jupyter notebook and perform all cleaning/preprocessing - Elliott
        - Provide ready-to-use data for DB population - Elliott
    - Infrastructure/Deployment:
        - EC2 setup - Assign if in scope
        - RDS setup - Assign if in scope